K.I.V.A.S.

STAGE 1 | TEAM PROJECT PLAN & MASTER TASK LIST

Module: Algorithms, Data Structures & Connectionist AI
Course: Foundations of Computer Science & Artificial Intelligence
Institution: Beijing Institute of Technology
Stage 1 deadline: 24 October 2026, 23:59 (Beijing time)

Document status: Initial working draft prepared for team review. Fields marked [TO CONFIRM] and statuses must be updated before submission. No unfinished work is represented as completed.

1. Team information and responsibilities

Member	Role	Strength / relevant experience	Development need / responsibility
Nihat [1820262201]	Coordinator (CEO)	Team coordination [confirmed]	Lead meetings, record decisions, maintain project log and progress
Huseyn [1820262217]	Content Lead	Programming and content organization [confirmed]	Research, sequence topics, draft/check Markdown learning content
Asad [1820262195]	Video Lead	Video production [confirmed]	Plan tutorial demonstrations and narration
Ali [1820262285]	Repository & Submission Lead (CIO)	Repository management [confirmed]	Set up GitHub, manage file structure and submit deliverables
Hasan [1820262209]	Video Lead	Video editing [confirmed]	Storyboard, recording and editing with Asad
Sadiq [1820262235]	Visual Design Lead	Visual design [confirmed]	Create diagrams, slide layouts and visual consistency
Mahmud [1820262247]	Quality Lead	Quality review [confirmed]	Check technical accuracy, readability, activities and evidence

All seven members will study the core module, author or review substantive material, document their own contributions, and participate in decisions. Roles assign leadership, not exclusive ownership. Full roster names, student IDs, actual strengths and skill gaps must be confirmed by each member.

2. Audience, learning goals and scope

Target learners: First-year university students with basic computer literacy and little or no prior knowledge of algorithms, data structures or neural networks. Simple arithmetic and the idea of variables are helpful; unfamiliar programming concepts will be explained.

By the end, learners should be able to:

• Describe an algorithm and trace linear search, binary search and a basic sorting procedure.

• Explain why sorted input is required for binary search and compare simple Big O growth rates.

• Identify arrays, linked lists, stacks, queues, trees and graphs; choose suitable structures for simple scenarios.

• Describe artificial neurons, weights, bias, activation functions and feedforward layers.

• Trace a small numerical neuron example and summarize prediction, loss and weight updating during training.

• Explain how data structures, algorithms and connectionist AI work together in applications.

Detailed coverage: (A) Algorithm fundamentals, pseudocode/flowcharts, searching, sorting, time complexity; (B) Arrays, linked lists, stacks, queues, trees, graphs and use-case comparisons; (C) Connectionist AI, artificial neurons, neural-network architecture, forward pass, introductory training, applications and limitations.

Depth boundaries: Explain linear/binary search and basic neuron arithmetic step by step; introduce Bubble Sort and Big O visually. Explain structures conceptually with examples. Mention backpropagation and gradient-based optimization at a conceptual level only. Advanced proofs, balanced trees, graph algorithm implementations, calculus-based derivations and large-scale model engineering are outside the first teaching version.

Open scope question for teacher: Confirm whether the course topic menu expects a narrower subset or a different emphasis across these three areas.

3. Planned connected learning resources

Resource	Structure and learning purpose
Shared Markdown	Single source of truth: section goals, definitions, clear explanations, diagrams, worked examples, practice activities, answers and cited references. Use relative links for assets.
Study Guide & Test Bank	Guide follows Markdown sections with concise recaps and practice. Test bank contains basic recall, tracing and application questions with answer keys and explanations.
PPT Slides	Approx. 30 editable 16:9 teaching slides, organized into introduction, algorithms, data structures, connectionist AI and review; diagrams and step-by-step demonstrations replace dense paragraphs.
Video Tutorials	Short narrated lessons: searching/sorting demonstration; data structure operations; artificial neuron and training loop. Scripts and visuals derived from checked Markdown.
Live Mini Lecture	Proposed question: “How can binary search find a value without checking every element?” Show a sorted array, have learners choose the next half, and compare against linear search. Confirm final topic with teacher.

A correction in Markdown triggers review of the matching Study Guide page, slide, video script, quiz item and references. All derived resources must stay consistent with the checked source.

4. Workflow, collaboration and quality control

• Research: divide subtopics, compare reputable textbooks/course materials, and write explanations in original beginner-friendly language.

• GitHub: shared repository with README, content/, assets/, slides/, videos/, project-plan/ and references/; use descriptive commits and review changes before merging.

• Review: a second member checks technical claims, examples, learner activities, answers, sources and broken image links. Mahmud coordinates the review checklist.

• Meetings: short coordination check-ins at least twice weekly and extra reviews before checkpoints; Nihat records date, attendees, decisions, actions and owners.

• Tracking: maintain a task table with owner, deliverable, internal due date, status, dependency and evidence link. Use honest statuses: Not started / In progress / Under review / Completed.

• AI and evidence: log significant AI assistance, prompts/outputs, independent verification and decisions; cite outside material. Each member keeps personal learning records.

• Risks: broad topic, unfamiliar Markdown/GitHub, inaccurate explanations, uneven workload, video delays. Mitigation: prioritize one complete topic, peer-review early, rebalance tasks and use internal deadlines.

5. Milestones and internal schedule

Date (Beijing)	Checkpoint
17-18 Oct	Confirm team information, module scope and GitHub access; assign detailed topics and begin dated logs.
19-20 Oct	Finish module outline; draft at least one complete Markdown topic and initial examples/diagrams.
21-22 Oct	Review content, test activities, collect actual GitHub contribution evidence; revise plan and logs.
23 Oct	Freeze Stage 1 snapshot, assemble PDF/ZIP evidence, check links and readability; all members review.
24 Oct	Submit Stage 1 plan, GitHub/Markdown V1.0, project log and combined personal logs by 23:59.
27-29 Oct	Build and rehearse the separate 4-minute status report using actual progress screenshots; test USB offline.
30 Oct	Submit/present Stage 1 Status Report by announced deadline/class schedule.
7 Nov	Stage 2: first versions of resources and mini-lecture work.
28 Nov	Stage 3: revised, checked final teaching resources and delivery.

6. Master task list — Stage 1 detailed

Task / expected result	Lead / support	Internal due	Status	Depends on
Confirm roster, IDs, strengths, gaps / verified team table	Nihat / all	18 Oct	In progress	Member replies
Agree scope and learner goals / approved outline	Huseyn / all	18 Oct	In progress	Module confirmation
Create repo, access, README / usable GitHub repository	Ali / all	18 Oct	In progress	Team GitHub usernames
Create section outline and statuses / outline.md	Huseyn / all	19 Oct	Not started	Scope agreement
Research algorithms and verify examples / cited notes	Huseyn + assigned member	20 Oct	Not started	Outline
Research structures / cited notes and diagrams	Sadiq + assigned member	20 Oct	Not started	Outline
Research connectionist AI / cited explanations	Huseyn + assigned member	20 Oct	Not started	Outline
Draft one complete coherent topic / .md with activity and answer	Huseyn / all	21 Oct	Not started	Research
Create visuals and check relative asset links / assets	Sadiq / Huseyn	21 Oct	Not started	Content draft
Peer review claims, examples, sources / review record	Mahmud / all	22 Oct	Not started	Markdown draft
Plan PPT, guide, videos and mini lecture / resource plan	Huseyn / Sadiq, Asan, Hasan	22 Oct	Not started	Checked outline
Capture GitHub history and meaningful changes / evidence	Ali / all	22 Oct	Not started	Real commits
Update meeting/project log and AI-use records / project log	Nihat / all	Daily to 23 Oct	In progress	Actual work
Write own course/project learning logs / seven personal logs	Each member	23 Oct	In progress	Own work/evidence
Assemble Stage 1 submission files / plan, report, ZIP, PDFs	Ali / all	23 Oct	Not started	Reviewed content
Final accuracy, offline and submission check / approved package	Mahmud + Ali / all	24 Oct	Not started	All deliverables

Status entries above are planning placeholders, not verified progress. Update them against real evidence before sharing or submitting.

7. Master task list — Stages 2 and 3

Task / expected result	Lead	Target	Status
Expand and review all Markdown chapters / checked content	Huseyn + all	5 Nov	Planned
Develop Study Guide and answer-keyed Test Bank / drafts	Huseyn + Mahmud	5 Nov	Planned
Produce teaching PPT with diagrams / editable slides	Sadiq + Huseyn	5 Nov	Planned
Record and edit tutorial videos / draft MP4s	Asad + Hasan	6 Nov	Planned
Prepare and rehearse mini lecture / timed lesson and activity	Nihat + all	Stage 2	Planned
Test learning activities with peers / feedback notes	Mahmud + all	Stage 2	Planned
Revise all resources consistently / corrected final versions	All; Huseyn coordinates	26 Nov	Planned
Verify sources, accessibility, offline assets and final package	Mahmud + Ali	27 Nov	Planned
Submit final package and update logs / delivery evidence	Ali + Nihat	28 Nov	Planned

8. Submission, GitHub and evidence

GitHub repository URL: https://github.com/aarzumanli/AI_project
Team name confirmation: K.I.V.A.S.
Plan filename: KIVAS_Stage1_ProjectPlan.pdf

The submission lead will submit one PDF or Word document for this plan after all members review it. The document should contain the current task-list snapshot and working repository link. Separate Stage 1 submissions include the GitHub/Markdown ZIP, project log, and combined private personal logs.

Evidence to maintain: dated meeting decisions, commits and actual diffs, reviewed content excerpts, verified example results, diagrams, relevant AI interactions and independent checks. Do not publish private student logs in a public repository.

9. AI-use declaration and review sign-off

This initial plan draft was prepared with assistance from ChatGPT on 17 October 2026 to organize the teacher-provided requirements, proposed workflow and task breakdown. The team must independently check every role, scope decision, schedule, progress status and dependency; record the relevant AI interaction and its verification in the project/personal logs. This document does not claim that team review or verification has already happened.

Review: Nihat [yes], Huseyn [yes], Asan [yes], Ali [yes], Hasan [yes], Sadiq [yes], Mahmud [yes]
Teacher feedback / scope decisions: [TO CONFIRM]